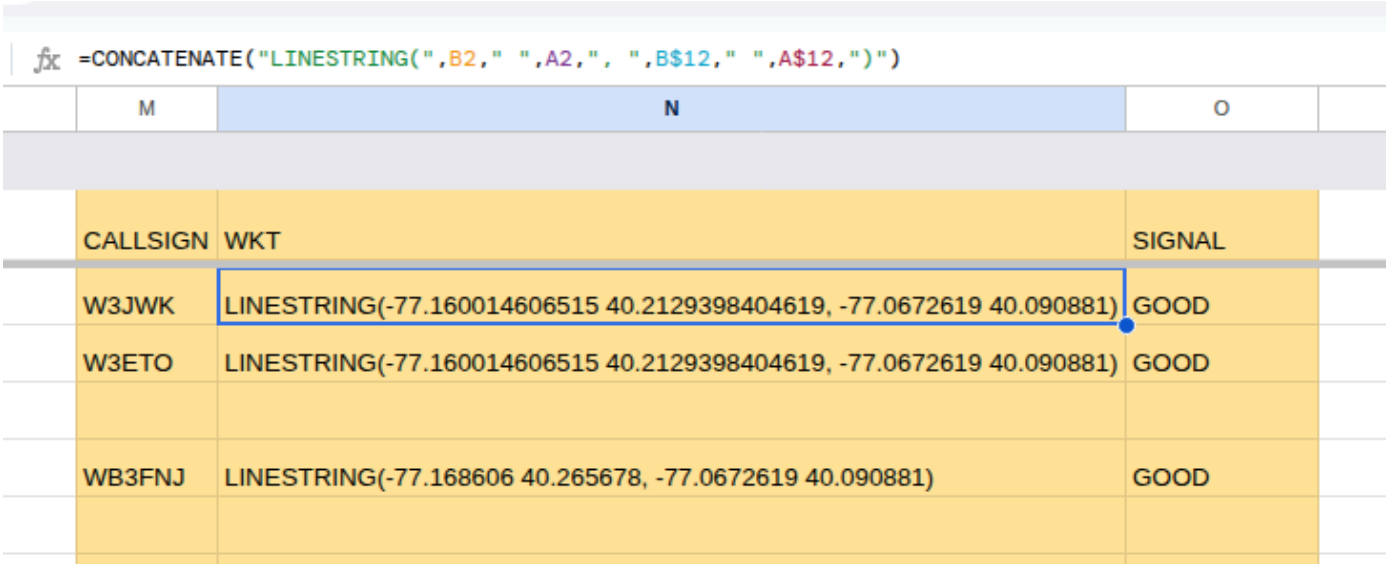
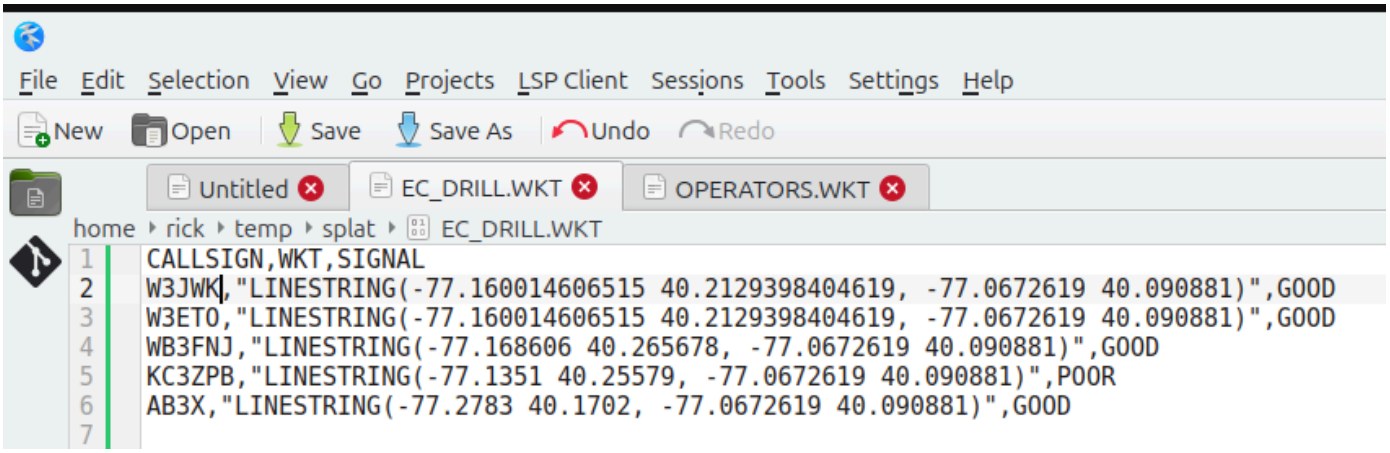
Point-to-Point Contact Map

Look at my spreadsheet. I added two sections with =CONCATENATE() functions to build strings called WKT - Well-Known-Text. WKT is a way to describe geo objects in text format and then can be imported and used by a geo program.

The section in yellow is defining a "LINESTRING" which will draw a line between two long/lat points. I also have the callsign to be used as the string's label and the signal column as a category to be used for assigning a color. I used my location for each one to represent what I was able to hear.

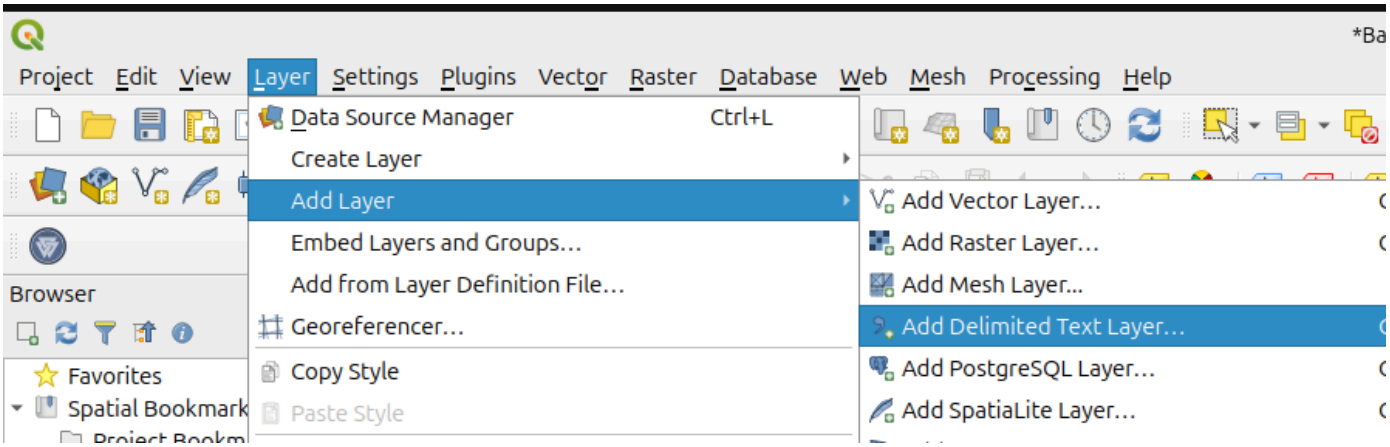


Those 3 columns have to be saved to a CSV file. For now, I just copied them and pasted it into a text file but then I had to add the commas between fields and the double quotes around the formula.

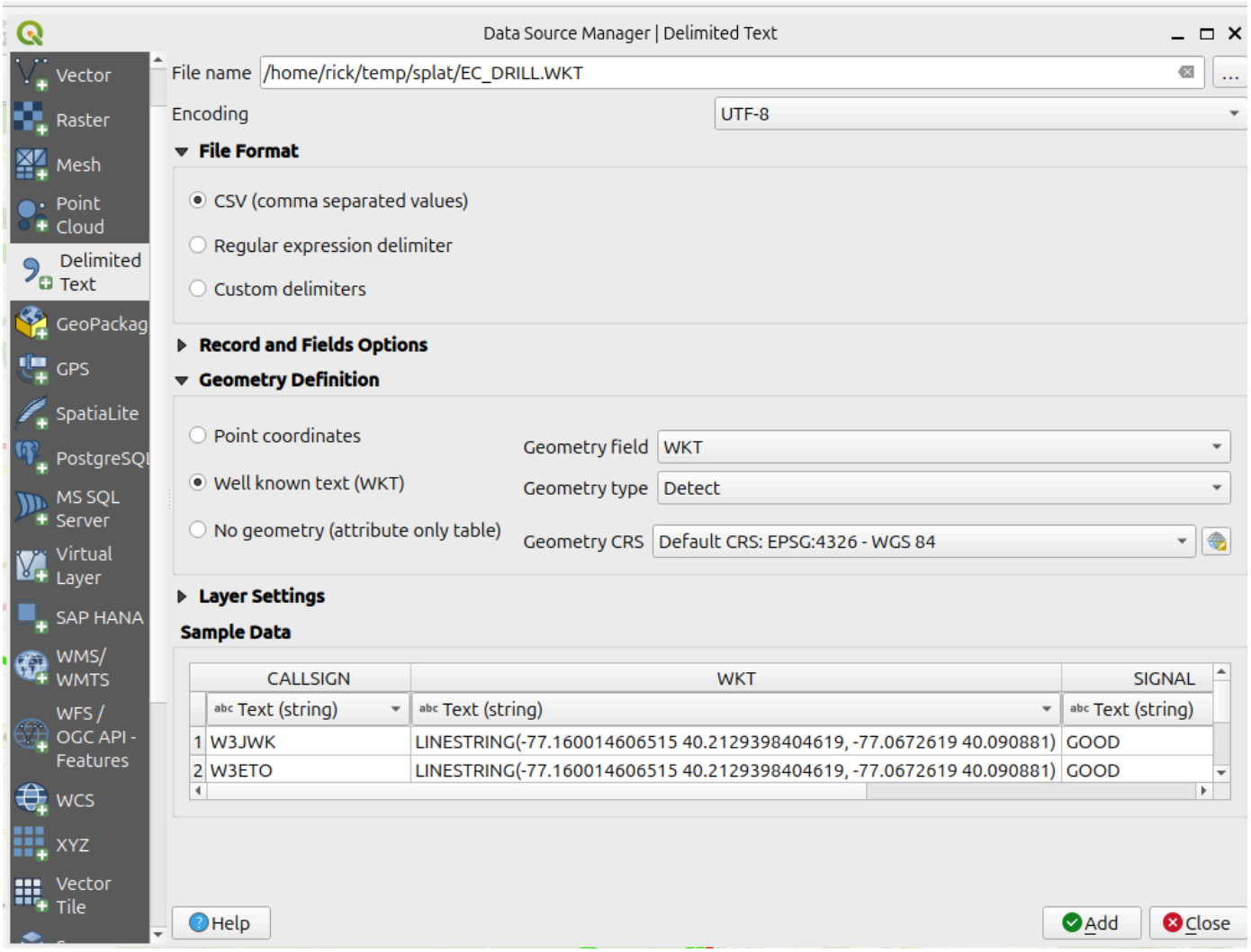


In QGIS, I started with a base map using OpenStreetMaps and set its transparency to 50%. This gives you a visual reference but makes it fainter so that the other layers stand out more.

Next I added a layer as a "delimited" layer.

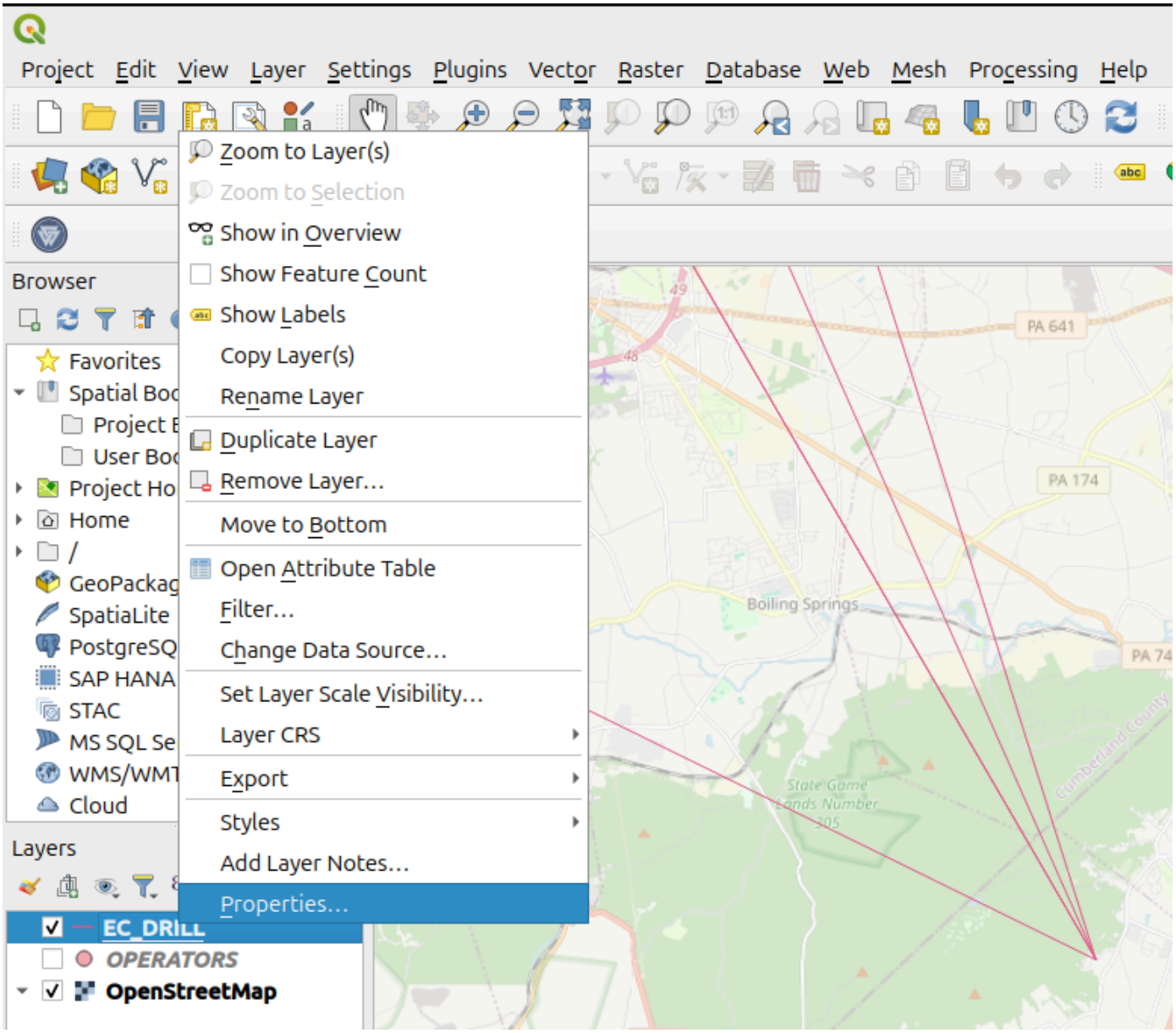


Next you need to set how to read the file. For file name, point to your input file. Under file format, set CSV. Under the Geometry Definition section, set the values as I show here. Then click OK.

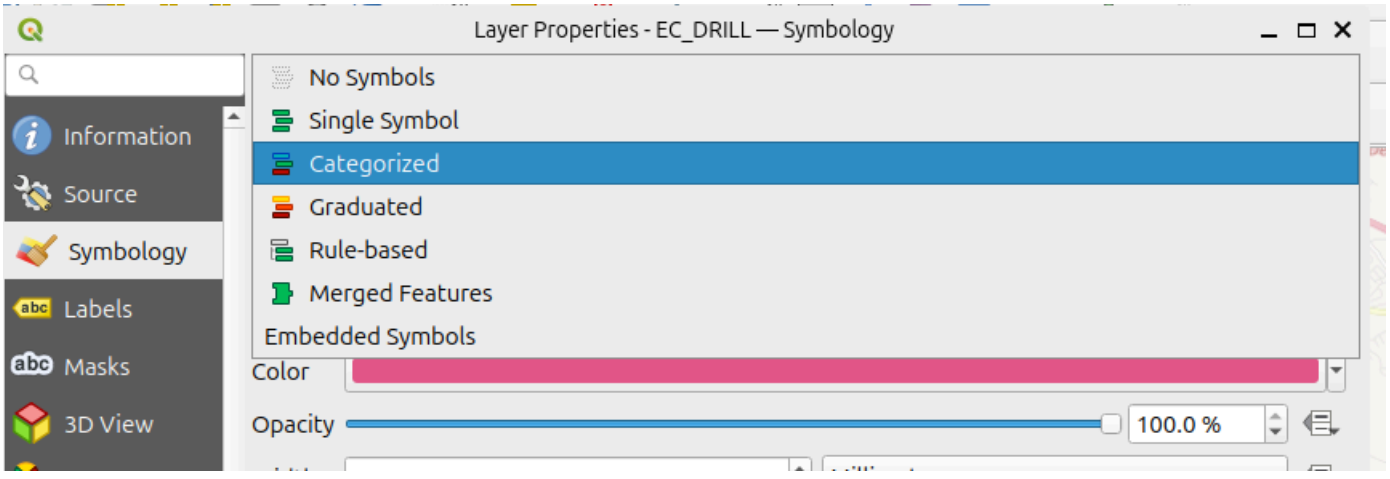


After you click OK, you should see a new layer in the layer side-panel "EC_Drill" and the lines showing up on the map. As shown here, they are all red for now.

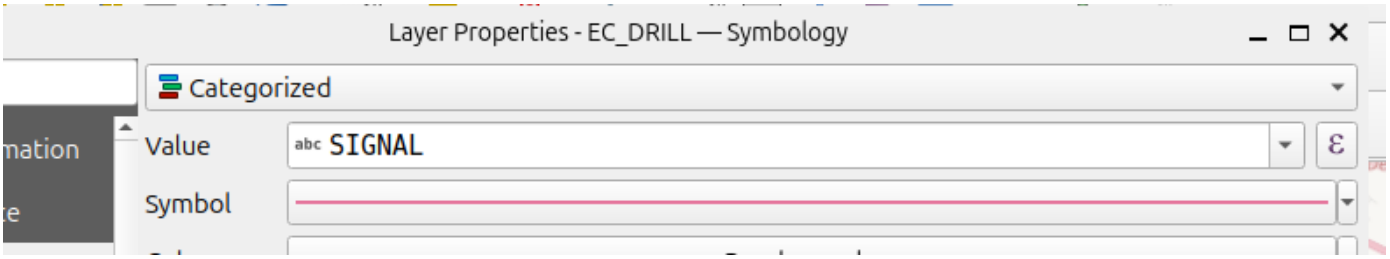
Right click on the layer name and pick "Properties".



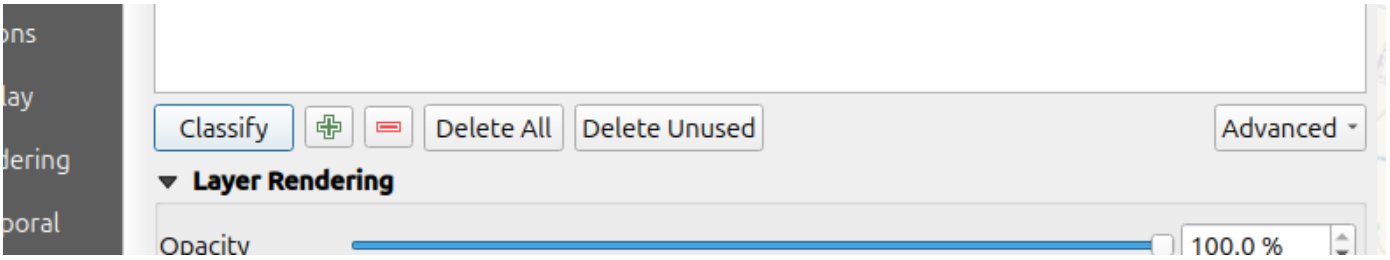
Click on "**Symbology**" and then change it from "Single Symbol" to "**Categorized**".



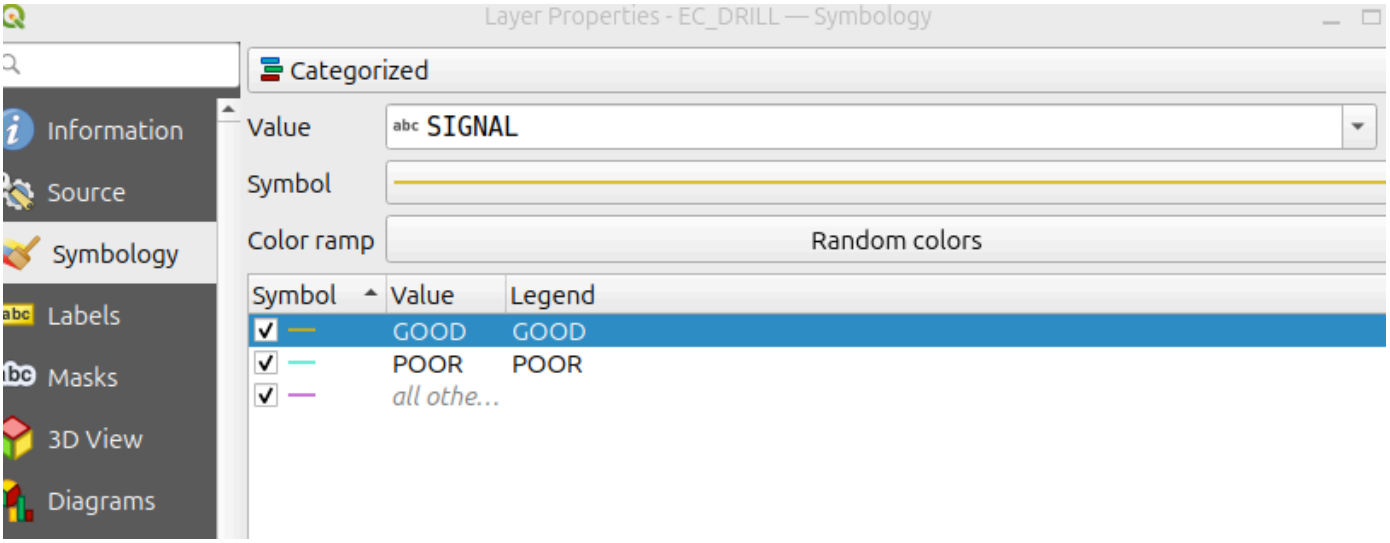
Change the "Value" field to point to the "Signal" column.



Click the "Classify" button to load the values from that column.

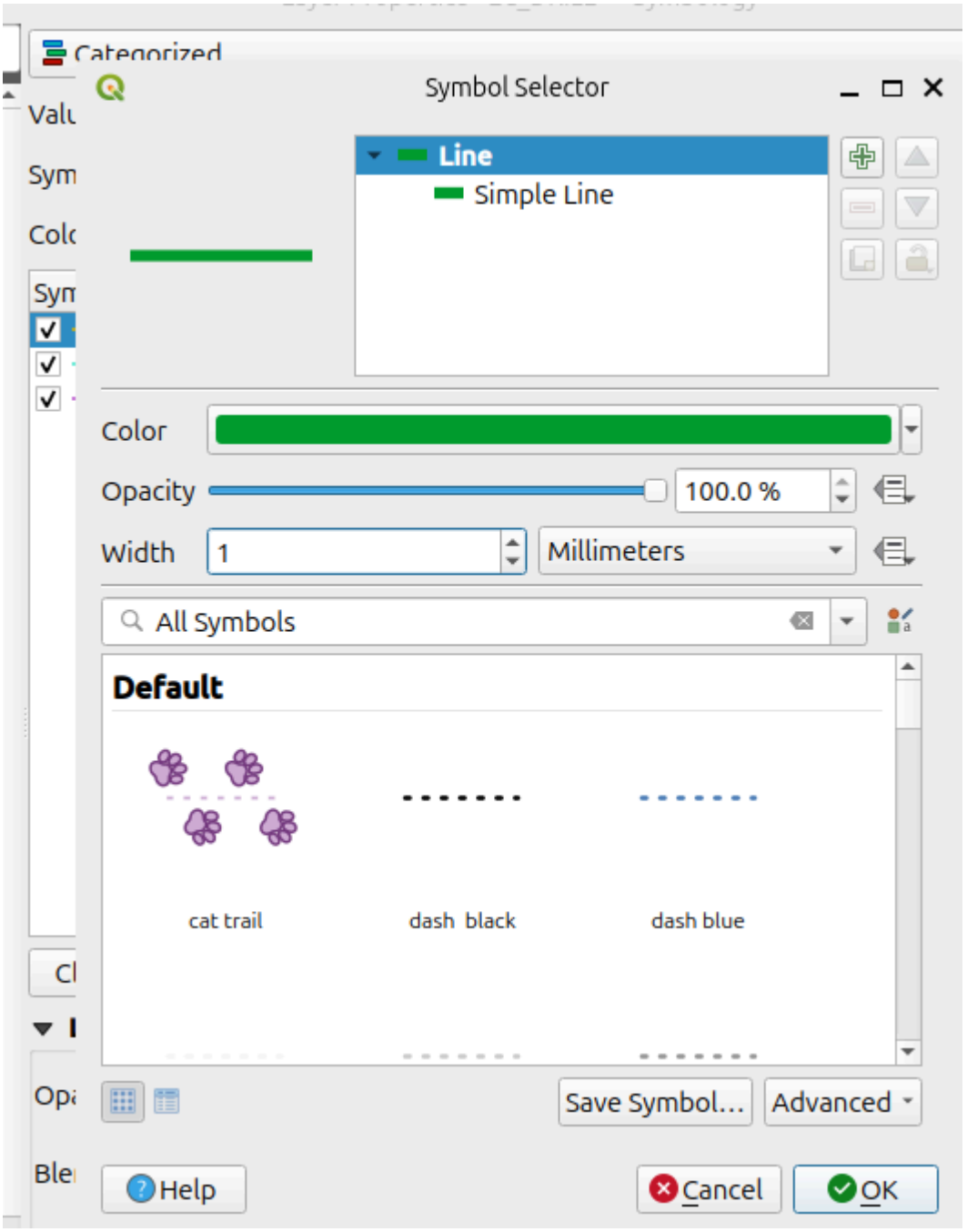


That will list all the unique values from the signal column.



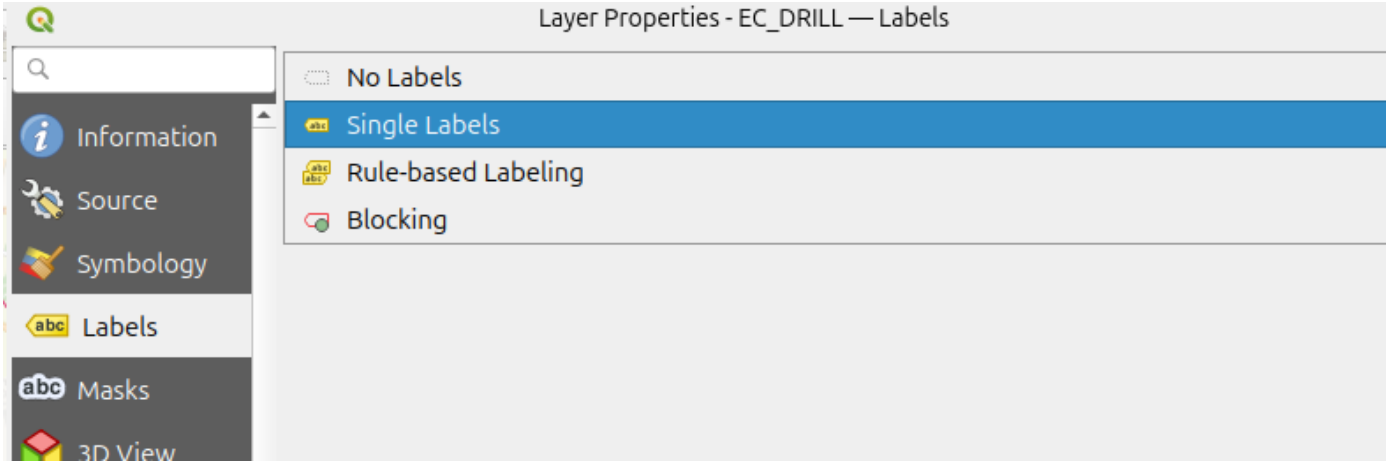
Double click on the assigned Symbol area of each entry to change the assigned color.

I changed "Good" to green and made the line 1mm wide to show up better. Do something similar for the "Poor" signals.

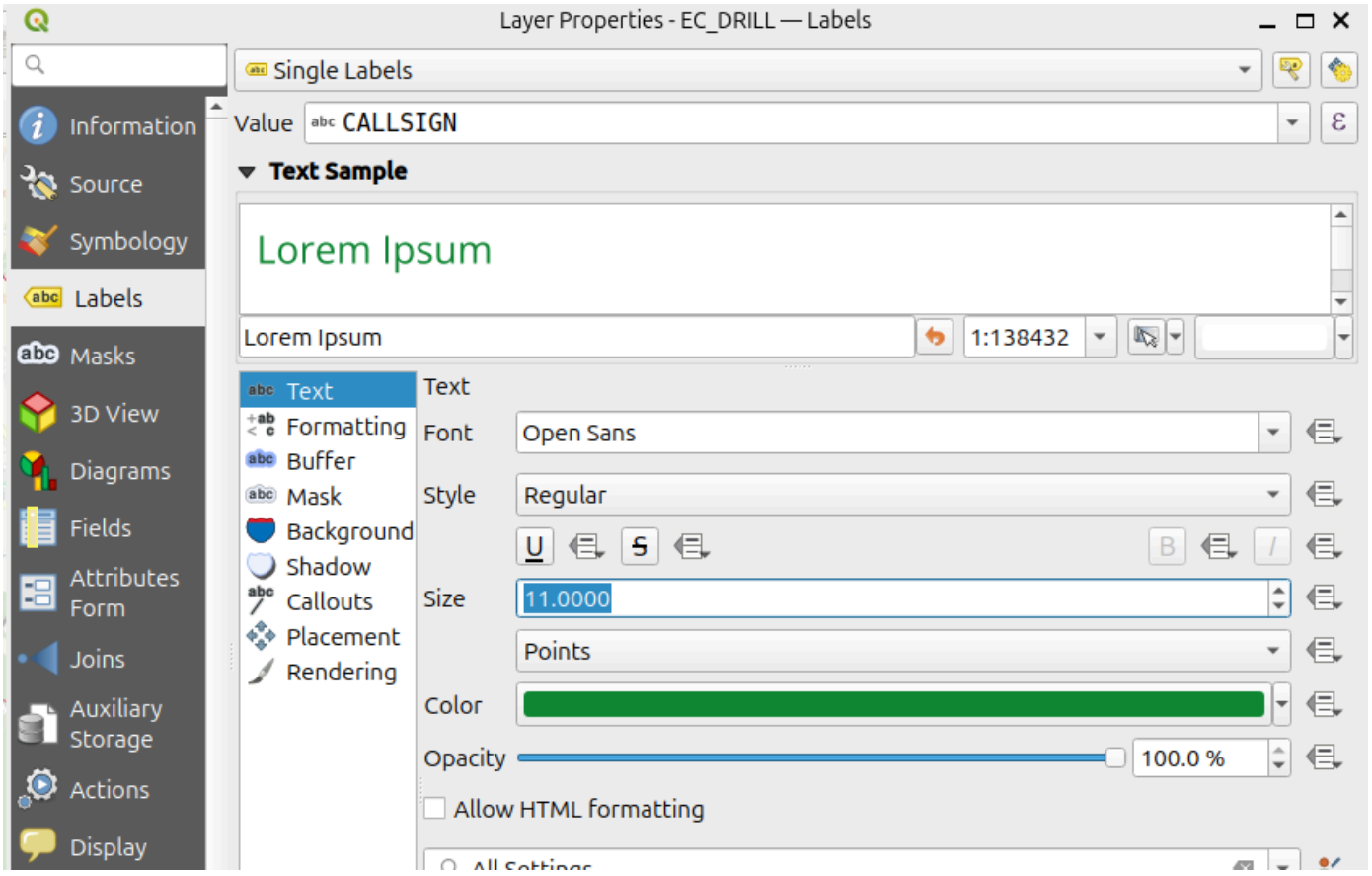


Next, click on the "Label" tab. This allows you to assign a label to the line.

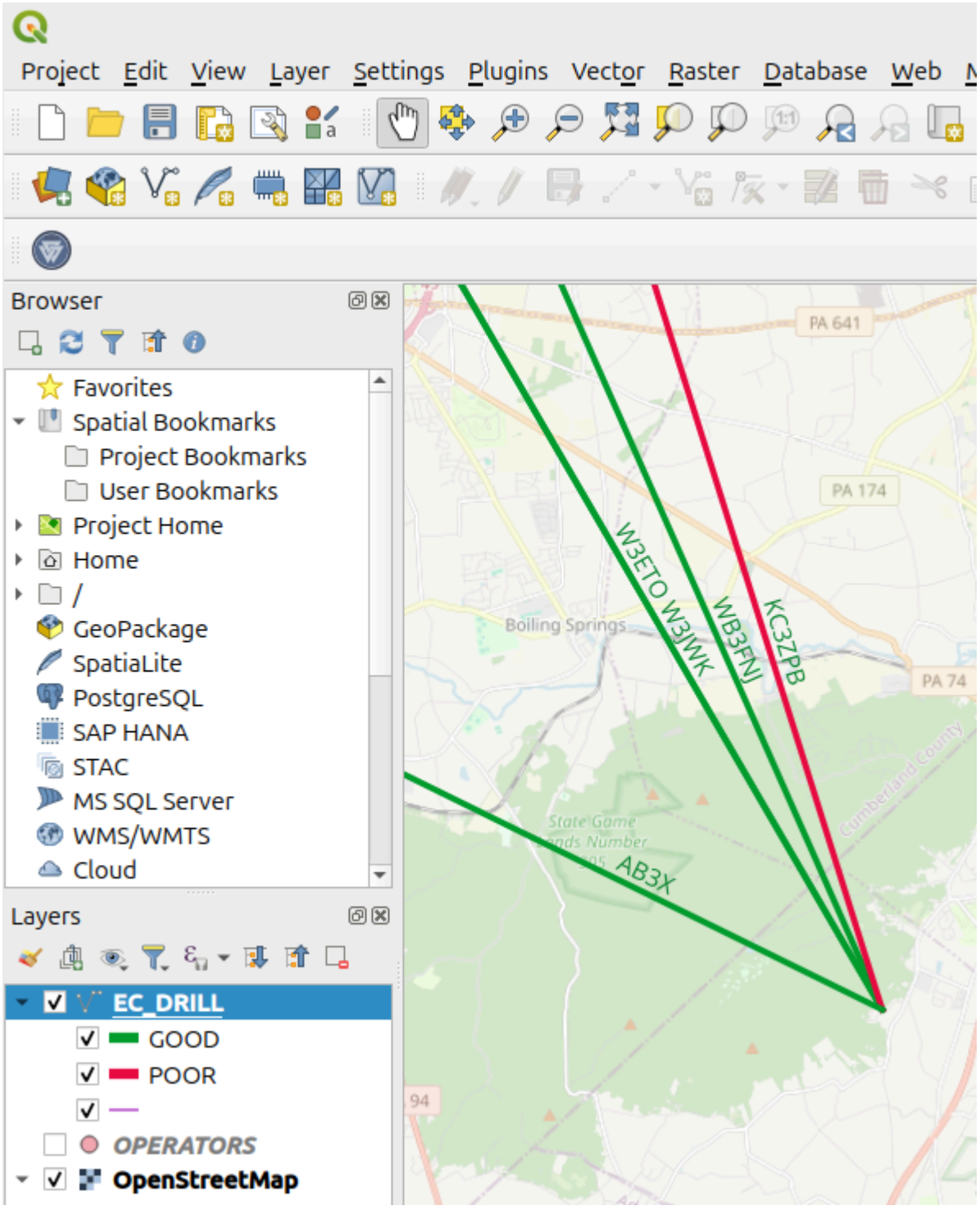
Change it from No Labels to "Single Labels".



In the "Value" field, point it to the CALLSIGN field. I change text color to green and increased the font size to 11. Then click OK.



You should now see the lines in red/green depending upon the signal report and they're labeled with the person I was hearing. In the Layer panel, you should see the fields and color assignments too.



I did a similar approach to get the dots on the map for operators. In the google sheet, I used WKT "POINT" to define a point with the operator's callsign.

CALLSIGN	WKT
W3JWK	POINT(-77.160014606515 40.2129398404619)
W3ETO	POINT(-77.160014606515 40.2129398404619)
WB3FNJ	POINT(-77.168606 40.265678)

The other steps are similar to what I walked through above. The final product should look something like this:

